# DIP-SMC-PSO: Technical Progress Memo

Model choices, algorithm design, and experimental results

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# 1 System Model: Equations and Parameter Choices

## 1.1 State Vector and Coordinate Definition

- **DIP state vector:**

$$\boldsymbol{x} = \left[x,\ \theta_1,\ \theta_2,\ \dot{x},\ \dot{\theta}_1,\ \dot{\theta}_2\right]^\top \in \mathbb{R}^6 \tag{1}$$

- **Coordinate convention:**

$$\boldsymbol{q} = [x,\ \theta_1,\ \theta_2]^\top, \qquad \dot{\boldsymbol{q}} = [\dot{x},\ \dot{\theta}_1,\ \dot{\theta}_2]^\top \tag{2}$$

  Units: $x$ [m], $\theta_i$ [rad], $\dot{x}$ [m/s], $\dot{\theta}_i$ [rad/s]. Sign convention: $+x$ rightward, $+\theta_i$ counterclockwise from upright.

- **Target equilibrium and error:**

$$\boldsymbol{x}^\star = [0,\ 0,\ 0,\ 0,\ 0,\ 0]^\top, \qquad \mathbf{e} = \boldsymbol{x} - \boldsymbol{x}^\star \tag{3}$$

  Tracking case: shift only the first component of $\boldsymbol{x}^\star$.

- **Measured output:**

$$\boldsymbol{y} = [x,\ \theta_1,\ \theta_2,\ \dot{x},\ \dot{\theta}_1,\ \dot{\theta}_2]^\top \tag{4}$$

  Feedback source: full-state in simulation; encoder/estimator pipeline in HIL. Safety limits: benchmark configuration files.

## 1.2 Equations of Motion

- **Full nonlinear dynamics:**

$$\mathbf{M}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\,\dot{\boldsymbol{q}} + \mathbf{G}(\boldsymbol{q}) + \boldsymbol{F}_{\text{fric}} = \boldsymbol{B}\,u \tag{5}$$

  with $\boldsymbol{q} = [x, \theta_1, \theta_2]^\top$, $\boldsymbol{B} = [1, 0, 0]^\top$, and actuator bounds $u \in [-150, 150]$ N.

- **Modeling assumptions:** planar rigid-body dynamics, ideal revolute joints, single-input bounded actuation, viscous + Coulomb friction, and disturbance injection through benchmark profiles (MT-8/LT-6).

- **Equivalent first-order form for control design:**

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + g(\boldsymbol{x})\,u + \boldsymbol{d}(t) \tag{6}$$

  Disturbance term: $\boldsymbol{d}(t) = 0$ (nominal), $\boldsymbol{d}(t) \neq 0$ (rejection tests).

- **Structural properties used in proofs:** Properties: $\mathbf{M}(\boldsymbol{q})(\boldsymbol{q})$ symmetric positive definite; $\dot{\mathbf{M}}(\boldsymbol{q}) - 2\mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})$ skew-symmetric on admissible configurations.

- **Model partition in this workflow:**
  - Full model: final benchmark reporting.
  - Simplified model: PSO loops (reduced coupling, no gyroscopic term).

### Inertia matrix $\mathbf{M}(\boldsymbol{q})$ (3×3, symmetric, configuration-dependent):

- Compact notation: $c_1 = \cos\theta_1$, $c_2 = \cos\theta_2$, $c_{12} = \cos(\theta_1 - \theta_2)$.

$$\mathbf{M}(\boldsymbol{q}) = \begin{bmatrix} M + m_1 + m_2 & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix} \tag{7}$$

- Matrix entries:

$$\begin{aligned} a_{12} &= (m_1 l_{c1} + m_2 l_1)c_1 + m_2 l_{c2} c_2 \\ a_{13} &= m_2 l_{c2} c_2 \\ a_{22} &= m_1 l_{c1}^2 + m_2(l_1^2 + l_{c2}^2) + I_1 + I_2 + 2m_2 l_1 l_{c2} c_{12} \\ a_{23} &= m_2 l_{c2}^2 + I_2 + m_2 l_1 l_{c2} c_{12} \\ a_{33} &= m_2 l_{c2}^2 + I_2 \end{aligned} \tag{8}$$

- Dominant coupling: $a_{22}$ term $2m_2 l_1 l_{c2} c_{12}$; grows with large relative-link deflection.

### Coriolis/centrifugal matrix $\mathbf{C}(\boldsymbol{q},\dot{\boldsymbol{q}})$ (non-zero entries):

- Computed from Christoffel symbols of (7); zero entries omitted.

$$\begin{aligned} C_{01} &= -(m_1 l_{c1} + m_2 l_1)\sin\theta_1\,\dot{\theta}_1 - m_2 l_{c2}\sin\theta_2\,\dot{\theta}_2 \\ C_{02} &= -m_2 l_{c2}\sin\theta_2\,\dot{\theta}_2 \\ C_{11} &= C_{12} = -m_2 l_1 l_{c2}\sin(\theta_1 - \theta_2)\,\dot{\theta}_2 \\ C_{21} &= +m_2 l_1 l_{c2}\sin(\theta_1 - \theta_2)\,\dot{\theta}_1 \end{aligned} \tag{9}$$

- Indices $(0, 1, 2)$ correspond to $(x, \theta_1, \theta_2)$.
- Gyroscopic augmentation: $g_c(\dot{\theta}_1 + \dot{\theta}_2)$ added to $C_{01}$ and $-C_{10}$; coefficient 0.01 (config-enabled).

### Gravity vector $\mathbf{G}(\boldsymbol{q})$:

$$\mathbf{G}(\boldsymbol{q}) = \begin{bmatrix} 0 \\ G_2 \\ G_3 \end{bmatrix}$$

$$\begin{aligned} G_2 &= -(m_1 l_{c1} + m_2 l_1) g \sin\theta_1 \\ G_3 &= -m_2 l_{c2}\, g \sin\theta_2 \end{aligned} \tag{10}$$

- $G_1 = 0$ because gravity is perpendicular to cart translation.

### Friction model (viscous + Coulomb):

$$F_{\text{fric},i} = -b_i\,\dot{q}_i - \mu_i\ \text{sign}(\dot{q}_i)\,\mathbb{1}_{|\dot{q}_i| > 10^{-6}} \tag{11}$$

- Friction scope: all DOFs $i \in \{x, \theta_1, \theta_2\}$.
- Viscous coefficients: $(0.2, 0.005, 0.004)$.
- Coulomb coefficients: from config; Monte Carlo variation ±10%.
- Dead zone: $|\dot{q}_i| > 10^{-6}$ to avoid sign discontinuity at rest.
- Simulation model: full viscous + Coulomb form (11).
- PSO speedup: reduced coupling (0.7–0.8) and no gyroscopic term.
- Linearization: numerical $(\mathbf{A}, \mathbf{B})$ at $\boldsymbol{q}^\star = \mathbf{0}, \dot{\boldsymbol{q}}^\star = \mathbf{0}$ with $\varepsilon = 10^{-8}$.
- Sampling and saturation: $\Delta t = 0.01$ s, $u \in [-150, 150]$ N.

### 1.3 Physical Parameters

| Parameter | Symbol | Value |
|---|---|---|
| Cart mass | $M$ | 1.5 kg |
| Link 1 mass | $m_1$ | 0.2 kg |
| Link 2 mass | $m_2$ | 0.15 kg |
| Link 1 length | $l_1$ | 0.4 m |
| Link 2 length | $l_2$ | 0.3 m |
| Link 1 COM distance | $l_{c1}$ | 0.2 m |
| Link 2 COM distance | $l_{c2}$ | 0.15 m |
| Link 1 inertia | $I_1$ | 0.0081 kg·m$^2$ |
| Link 2 inertia | $I_2$ | 0.0034 kg·m$^2$ |
| Gravity | $g$ | 9.81 m/s$^2$ |
| Cart friction | $b_c$ | 0.2 N·s/m |
| Joint 1 friction | $b_1$ | 0.005 N·m·s/rad |
| Joint 2 friction | $b_2$ | 0.004 N·m·s/rad |
| Max control force | $u_{\max}$ | 150 N |
| Control period | $\Delta t$ | 0.01 s (100 Hz) |

### 1.4 Model Selection: Simplified vs. Full Nonlinear

| Model | Use in this project | Rationale |
|---|---|---|
| Simplified (linearized around upright) | PSO optimization (all controllers) | 10–50× faster per simulation step. PSO requires thousands of evaluations; full model would be prohibitive. |
| Full nonlinear (5) | All published benchmark results | Complete Lagrangian, all coupling terms, Coriolis, centrifugal, gyroscopic, full friction. Used for MT-5, MT-7, MT-8, LT-6. |
| Low-rank (SVD approximation) | HIL mode (hardware-in-the-loop) | Reduced singular-value truncation preserves stability with lower real-time compute load. |

- **Decision basis:** full nonlinear PSO ($40 \times 200 \times 10$ s) exceeds 8 h/controller.
- Optimization runtime: minutes on simplified model.
- Reporting model: full nonlinear only.

## 2 Sliding Mode Control: Surface Design and Control Laws

### 2.1 Sliding Surface

- **Compact state form:**

$$z = [\theta_1,\ \theta_2,\ \dot\theta_1,\ \dot\theta_2]^\top, \qquad s = [\lambda_1,\ \lambda_2,\ k_1,\ k_2]^\top, \tag{12}$$

- **Equivalent error form** ($e_i = \theta_i - \theta_i^\star$, $\theta_i^\star = 0$):

$$\sigma = \lambda_1 e_1 + \lambda_2 e_2 + k_1 \dot e_1 + k_2 \dot e_2 \tag{13}$$

- **Sliding dynamics and stability condition:**

$$\sigma = 0 \ \Rightarrow\ \dot e_i = -\frac{\lambda_i}{k_i} e_i, \quad \lambda_i > 0,\ k_i > 0,\ i \in \{1,2\} \tag{14}$$

Time constants on surface: $\tau_i = k_i/\lambda_i$ (direct tuning meaning).

- **Numerical tuning example (MT-8 robust Classical gains):** $(\lambda_1, \lambda_2, k_1, k_2) = (5.51, 3.49, 23.07, 12.85)$ gives $\tau_1 \approx 4.19$ s and $\tau_2 \approx 3.68$ s.
- **Units and scaling:** Signal units: $\theta_i$ [rad], $\dot\theta_i$ [rad/s]. Scaling policy: no state normalization before PSO; scaling handled by gain bounds and cost normalization.
- **Relative degree and robustness channel (one-line form):**

$$\dot\sigma = \phi(\boldsymbol{x}, t) + b(\boldsymbol{x})u + b(\boldsymbol{x})d(t), \qquad b(\boldsymbol{x}) = \mathbf{L}\mathbf{M}^{-1}\mathbf{B}, \quad \mathbf{L} = [0,\ k_1,\ k_2 \tag{15}$$

Reachability margin: if $|d(t)| \le \bar d$, select $K > \bar d + \eta$, $\eta > 0$.

- **Practical limitation and mitigation:** Limitation: surface excludes cart position $x$; aggressive recovery may increase cart excursion. Mitigation set: hard saturation $u \in [-150, 150]$ N, benchmark scenario bounds, optional outer-loop cart reference shaping.

### 2.2 Controller Variants: Control Laws

- **Unified template (all variants):**

$$u = u_{\text{eq}} + u_{\text{sw}} + u_{\text{aux}} \tag{16}$$

Mapping: variant-specific choices of $(u_{\text{eq}}, u_{\text{sw}}, u_{\text{aux}})$.

#### 2.2.1 *Classical SMC: Saturation Function and Equivalent Control*

- **Saturation function:**

$$\operatorname{sat}\left(\frac{\sigma}{\varepsilon}\right) = \begin{cases} \sigma/\varepsilon & |\sigma| \le \varepsilon \\ \operatorname{sign}(\sigma) & |\sigma| > \varepsilon \end{cases} \tag{17}$$

- **Control law mapping to** (16)**:**

$$u = \underbrace{u_{\text{eq}}}_{\text{equivalent}} + \underbrace{K \operatorname{sat}\left(\frac{\sigma}{\varepsilon}\right)}_{\text{switching}} + \underbrace{-k_d\,\dot\sigma}_{\text{auxiliary damping}} \tag{18}$$

- **Equivalent control term (from $\dot\sigma = 0$):**

$$u_{\text{eq}} = -\frac{\mathbf{L}\mathbf{M}^{-1}\mathbf{F}}{\mathbf{L}\mathbf{M}^{-1}\mathbf{B}}, \qquad \mathbf{F} = \mathbf{C}\dot{\boldsymbol{q}} + \mathbf{G} + \mathbf{F}_{\text{fric}} \tag{19}$$

- **Discrete-time implementation note ($T_s$ = 0.01 s):**

$$\dot\sigma[k] = \beta_f \dot\sigma[k-1] + (1-\beta_f)\frac{\sigma[k] - \sigma[k-1]}{T_s} \tag{20}$$

Sequence: compute $u_{\text{raw}}$ via (18), then clamp $u[k] = \operatorname{sat}_{[-150,150]}(u_{\text{raw}}[k])$.

#### 2.2.2 *Super-Twisting Algorithm (STA-SMC)*

- **Control decomposition:**

$$u = u_1 + u_2 \tag{21}$$

$$u_1 = -K_1\sqrt{|\sigma| + \delta}\ \operatorname{sign}(\sigma) \qquad \text{(continuous)}$$

$$\dot u_2 = -K_2\ \operatorname{sign}(\sigma) \qquad \text{(discontinuous derivative)}$$

Regularization: $\delta = 10^{-10}$ near $\sigma = 0$.

- **Integral anti-windup (implementation):**

$$u_2[k] = \operatorname{clip}\left(u_2[k-1] - K_2 T_s \operatorname{sign}(\sigma[k]),\ u_{2,\min},\ u_{2,\max}\right) \tag{22}$$

- **Gain condition and consistency note:** Required condition: $K_1 > K_2 > 0$. Flagged record: $(K_1, K_2) = (2.02, 6.67)$ violates condition; needs re-tuning.

#### 2.2.3 *Adaptive SMC*

- **Control law:**

$$u = -K(t)\ \mathrm{sat}\left(\frac{\sigma}{\varepsilon}\right) \tag{23}$$

- **Adaptive gain update with projection and dead zone:**

$$\dot{K}(t) = \Pi_{[K_{\min},K_{\max}]}(\gamma|\sigma| - \delta_{\text{leak}}K(t)), \quad |\sigma| \geq d_z \tag{24}$$

Dead-zone rule: $\dot{K}(t) = 0$ for $|\sigma| < d_z$.

- **Practical settings:** Bounds: $K(t) \in [0.1, 100]$; dead zone: $d_z = 0.05$; leak: $\delta_{\text{leak}} = 0.01$. Effect: prevents gain drift and noise-driven amplification.

#### 2.2.4 *Hybrid Adaptive STA-SMC*

- **Control law:**

$$u = -k_1(t)\,|\sigma|^{0.5}\,\mathrm{sign}(\sigma) - k_2(t)\int \mathrm{sign}(\sigma)\,dt \tag{25}$$

Gain rates: $\gamma_1 = 0.1$, $\gamma_2 = 0.05$; rate limiter: 5.0/s.

**Limitation (Hybrid Adaptive STA-SMC):** Observed instability at large disturbances: up to 666.9° overshoot and 176% chattering increase. Status: retained for completeness; **not recommended for deployment**.

### 2.2 Comparison Snapshot

| Variant | Control continuity | # gains | Chattering trend | Compute load | Deployment |
|---|---|---|---|---|---|
| Classical SMC | Piecewise continuous (`sat`) | 6 | Medium | Low | Recommended baseline |
| STA-SMC | Continuous $u$ (with bounded $u_2$) | 6 | Low | Medium | Recommended |
| Adaptive SMC | Piecewise continuous + adaptive gain | 5 | Low–medium | High | Recommended (uncertainty-heavy) |
| Hybrid Adaptive STA | Continuous + adaptive | 4 | Unstable in tests | Medium–high | Not recommended |

### Implementation Order (all variants)

- Compute $\sigma$ from current state and selected surface gains.
- Compute controller terms $(u_{\text{eq}}, u_{\text{sw}}, u_{\text{aux}})$.
- Apply actuator saturation: $u \leftarrow \mathrm{sat}_{[-150,150]}(u)$.
- Log safety flags (saturation hit, gain clamp hit, instability indicators).

## 3 PSO Algorithm: Configuration and Methodology

### 3.1 Algorithm Parameters

| Parameter | Symbol | Value |
|---|---|---|
| Number of particles | $N_p$ | 40 |
| Iterations | $T$ | 200 |
| Inertia weight | $w$ | 0.7 |
| Cognitive acceleration | $c_1$ | 2.0 |
| Social acceleration | $c_2$ | 2.0 |
| Random seed | – | 42 |

- **Coverage choice:** $N_p = 40$ for 6D search.
- Convergence budget: $T = 200$ iterations.
- Coefficients: $(w, c_1, c_2) = (0.7, 2.0, 2.0)$.
- Algorithm block (implementation order): Figure 1.
- Convergence evidence (real run): Figure 2.

| # | **Robust PSO tuning loop (code-level order)** |
|---|---|
| 1 | Initialize particles $(x_i, v_i)$ and set $p_i^\star \leftarrow x_i$. |
| 2 | For each particle, simulate all $N_s = 15$ scenarios and compute $J_i$. |
| 3 | Update personal best $p_i^\star$ and global best $g^\star$. |
| 4 | Update velocity and position using $(w, c_1, c_2)$ and random terms. |
| 5 | Apply gain-bound projection and STA constraint checks $(K_1 > K_2)$. |
| 6 | Repeat until $t = T$ (or early-stop criterion), then return $g^\star$. |

Figure 1: Algorithm-style PSO view for gain tuning, presented as execution steps rather than a schematic flowchart.

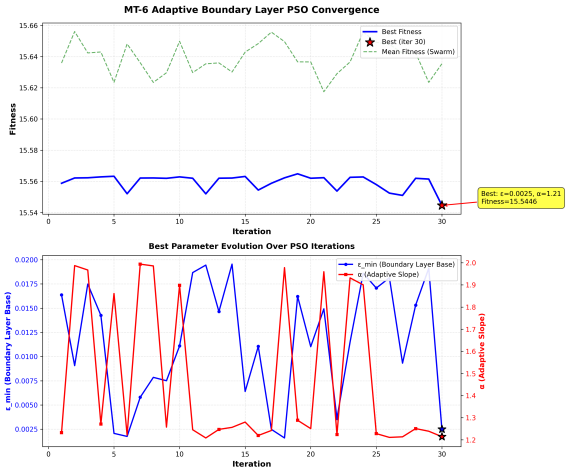


Figure 2: PSO convergence evidence from experiment logs (MT-6): best-fitness trend, swarm-mean fitness, and parameter evolution over iterations.

$$v_{i,d}^{t+1} = w\,v_{i,d}^t + c_1 r_1 (p_{i,d} - x_{i,d}^t) + c_2 r_2 (g_d - x_{i,d}^t), \qquad r_1, r_2 \sim U(0,1). \tag{26}$$

### 3.2 Fitness Function

$$J(\boldsymbol{\theta}) = w_1 \bar{J}_{\text{mc}} + w_2 \max_k J_k \tag{27}$$

- $w_1 = 0.7$, $w_2 = 0.3$ (mean vs. worst-case balance).

$$J_k = w_{\text{err}}\,\mathrm{ISE}(\boldsymbol{x}) + w_u\,\mathrm{IAE}(u) + w_r\,\mathrm{IAE}(\dot{u}) + w_s \int \sigma^2 dt + P \cdot \mathbb{1}_{\text{unstable}} \tag{28}$$

- Cost channels: tracking, effort, slew-rate, sliding-energy, instability.
- Weights: $[w_{\text{err}}, w_u, w_r, w_s] = [1.0, 0.1, 0.01, 0.1]$.
- Normalization thresholds: $[10.0, 100.0, 1000.0, 1.0]$.
- Instability penalty: $P = 10^6$.

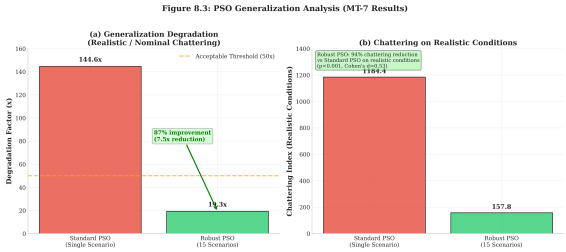


Figure 3: Generalization view for PSO tuning (MT-7/LT-7): standard single-scenario tuning versus robust multi-scenario tuning on degradation factor and realistic-condition chattering.

### 3.3 Search Bounds per Controller

| Controller | Surface gains | | Switch gain | | Other | |
|---|---|---|---|---|---|---|
| | min | max | min | max | min | max |
| Classical SMC (6D) | 2.0 | 30.0 | 2.0 | 50.0 | 0.05 | 3.0 |
| STA-SMC (6D) | 2.0 | 30.0 | 2.0 | 30.0 | 0.05 | 3.0 |
| Adaptive SMC (5D) | 2.0 | 30.0 | 2.0 | 10.0 | 0.05 | 3.0 |
| Hybrid Adaptive (4D) | 2.0 | 30.0 | 2.0 | 10.0 | – | – |

- STA constraint: $K_1 > K_2$.
- Violation penalty: $P = 10^6$.
- **Hyperparameter sensitivity (finding #16, #17):** no formal ablation over $N_p, w$; literature defaults used.
- Population heuristic: $N_p \approx 10d$; with $d = 6$, selected $N_p = 40$.
- Convergence logs (`academic/logs/pso/`): plateau at 150–180; budget fixed at 200 (no early stop).
- **Cost scale calibration (finding #18):** thresholds $[10, 100, 1000, 1]$ reflect unstable-run magnitudes.
- Penalty magnitude: $P = 10^6 \approx 10^4\times$ stable cost.

### 3.4 Robust PSO: Scenario-Based Optimization

- Failure mode: nominal-only PSO under varied initial conditions.

$$2.14 \pm 0.13 \longrightarrow 107.61 \pm 5.48 \quad (\times 50.4) \tag{29}$$

- **Robust PSO rule:** evaluate each particle on $N_s = 15$ scenarios.
  - 3 nominal: $\theta_0 \in [-0.05, +0.05]$ rad
  - 4 moderate: $\theta_0 \in [-0.15, +0.15]$ rad
  - 8 large: $\theta_0 \in [-0.30, +0.30]$ rad
- Fitness: $J_{\text{robust}} = 0.7\,\bar{J} + 0.3 \max J_k$.
- Improvement: $50.4\times \rightarrow 7.5\times$ ($144.59\times \rightarrow 19.28\times$).
- Before/after generalization view: Figure 3.
- **Scenario split rationale (finding #19):** 3+4+8 (20%, 27%, 53%); large-perturbation emphasis.
- Missing analysis: no split-sensitivity ablation.

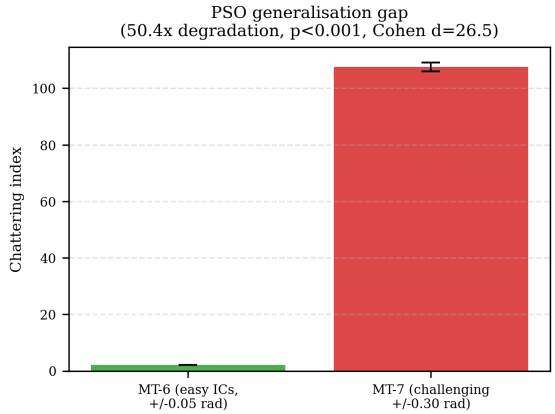


Figure 4: Chattering index: easy initial conditions (MT-6, ±0.05 rad) versus challenging conditions (MT-7, ±0.30 rad). The 50.4× degradation (Cohen $d = 26.5$, $p < 0.001$) motivates robust multi-scenario PSO tuning.

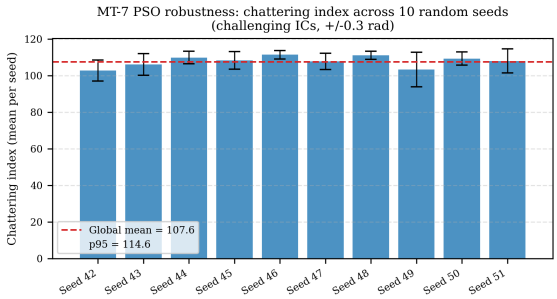


Figure 5: MT-7 PSO reproducibility: chattering index (mean ± SD) for 10 random seeds, each running 5 trials under challenging initial conditions. Global mean = 107.6 with CV = 5.1%, confirming low inter-seed variance.

### 3.5 PSO-Optimized Gains (MT-8 Robust Tuning)

| Controller | Nominal gains | Robust gains |
|---|---|---|
| Classical SMC | k1=5.0, k2=5.0, l1=5.0, l2=0.5, K=0.5, kd=0.5 | k1=23.07, k2=12.85, l1=5.51, l2=3.49, K=2.23, kd=0.15 |
| STA-SMC | c1=12.0, l1=6.0, c2=4.85, l2=3.43, K1=8.0, K2=4.0 | c1=5.62, l1=3.75, c2=4.36, l2=2.05, K1=2.02, K2=6.67 |
| Adaptive SMC | k1=10.0, k2=8.0, l1=5.0, l2=4.0, gamma=1.0 | k1=2.14, k2=3.36, l1=7.20, l2=0.34, gamma=0.29 |
| Hybrid Adaptive STA | c1=5, l1=5, c2=5, l2=0.5 | c1=10.15, l1=12.84, c2=6.82, l2=2.75 |

- Cost deltas (robust vs nominal): Classical $-2.2\%$, Hybrid $-21.4\%$.
- STA/Adaptive rows: consistency-check references.
- **Known inconsistency (finding #20):** robust STA gains ($K_1 = 2.02$, $K_2 = 6.67$) violate $K_1 > K_2$.
- Guarantee condition (Moreno–Osorio): $K_1 > K_2 > 0$.
- Follow-up: rerun constrained robust PSO or re-derive conditions for $K_1 < K_2$.

## 4 Lyapunov Stability: Proofs and Validation

- Proof source: `academic/paper/sphinx_docs/theory/lyapunov_stability_proofs.md` (v2.0, 1427 lines, 9 theorems, 5 lemmas, 15 references).
- **Symbolic constants note (finding #21):** $\beta, \eta, c_1, c_2, \alpha_1, \alpha_2$ are design/state-dependent.
- $\beta = \mathbf{L}\mathbf{M}^{-1}(\boldsymbol{q})\mathbf{B} > 0$; configuration-dependent.
- $\eta = K - \bar{d} > 0$; design-enforced.
- Proof objective: positivity and boundedness along trajectories.

### 4.1 Classical SMC

$$V = \tfrac{1}{2}\sigma^2 \tag{30}$$

$$\dot{V} = \sigma\dot{\sigma} \le -\beta\eta|\sigma| - \beta k_d \sigma^2 < 0 \tag{31}$$

- Valid region: $|\sigma| > \varepsilon$ with $\beta = \mathbf{L}\mathbf{M}^{-1}\mathbf{B} > 0$, $\eta = K - \bar{d} > 0$.
- Reaching bound: $t_{\text{reach}} \le |\sigma(0)|/(\eta\beta)$.
- Stability type: **exponential asymptotic** ($k_d > 0$).
- Numerical check: 96.2% of samples satisfy $\dot{V} < 0$ outside boundary layer.
- Residual 3.8%: early reaching transitions (expected).

### 4.2 Super-Twisting (STA-SMC)

- Lyapunov form: generalized-gradient (Moreno & Osorio, 2008).

$$V_{\text{STA}} = |\sigma| + \frac{1}{2K_2}z^2, \qquad z = u_2 \tag{32}$$

$$\dot{V}_{\text{STA}} \le -c_1\|\boldsymbol{\xi}\|^{3/2} + c_2 L \tag{33}$$

- $\boldsymbol{\xi} = [|\sigma|^{1/2}\,\text{sign}(\sigma), z]^\top$.
- $L$ is the Lipschitz bound of $\dot{d}_u(t)$; $c_1, c_2 > 0$ depend on gains.
- **Moreno-Osorio conditions** (finite-time requirement):

$$K_1 > 2\sqrt{\frac{2\bar{d}}{\beta}}, \qquad K_2 > \frac{\bar{d}}{\beta}, \qquad K_1 > K_2 \tag{34}$$

- Disturbance bound: $\bar{d}$; channel gain: $\beta = \mathbf{L}\mathbf{M}^{-1}\mathbf{B}$.
- Nominal gains satisfy conditions ($K_1 = 8 > K_2 = 4$); robust gains do not.
- Convergence target: $\sigma(t) = 0$ for $t \ge T_f$.
- Closed-form DIP bound for $T_f$: not derived ($\beta$ is configuration-dependent).
- Stability type: **finite-time convergence**.
- Numerical check: 1.82 s convergence (16% faster than classical).

### 4.3 Adaptive SMC

- Composite Lyapunov function (state + parameter error):

$$V_{\text{ad}} = \tfrac{1}{2}\sigma^2 + \frac{1}{2\gamma}\tilde{K}^2, \qquad \tilde{K} = K(t) - K^* \tag{35}$$

$$\dot{V}_{\text{ad}} \le -\eta|\sigma| \quad \Rightarrow \quad \sigma(t) \to 0, \quad K(t) \text{ bounded} \tag{36}$$

- Stability type: **asymptotic with bounded adaptive gain**.
- Numerical check: 100% samples satisfy $K(t) \in [K_{\min}, K_{\max}] = [0.1, 100]$.

### 4.4 Hybrid Adaptive STA-SMC

- Extended composite Lyapunov:

$$V_{\text{hyb}} = \tfrac{1}{2}\sigma^2 + \frac{\tilde{k}_1^2}{2\gamma_1} + \frac{\tilde{k}_2^2}{2\gamma_2} + \tfrac{1}{2}u_{\text{int}}^2 \tag{37}$$

$$\dot{V}_{\text{hyb}} \le -\alpha_1 V_{\text{hyb}} + \alpha_2\|\boldsymbol{w}\| \tag{38}$$

- Stability type: **Input-to-State Stable (ISS)**.
- Numerical check: bounded signals; 0 emergency resets (nominal tests).

### 4.5 Swing-Up SMC

- Switched Lyapunov approach for two-phase control:

$$V_{\text{sw}}(t) = \begin{cases} E_{\text{total}} - E_{\text{upright}} & \text{(swing-up phase)} \\ \tfrac{1}{2}\sigma^2 & \text{(stabilization phase)} \end{cases} \tag{39}$$

- Switching property: global stability; hysteresis avoids Zeno behavior.

### 4.6 MPC

- Lyapunov candidate is optimal cost-to-go $V_k = J_k^*$:

$$V_{k+1} - V_k \le -\lambda_{\min}(\mathbf{Q})\|\boldsymbol{x}_k\|^2 < 0 \tag{40}$$

- Stability type: **asymptotic** (requires recursive feasibility).

### 4.7 Summary Table

| Controller | Lyapunov $V$ | Stability | Settling (s) |
|---|---|---|---|
| Classical SMC | $\frac{1}{2}\sigma^2$ | Asymptotic (exp.) | 2.15 |
| STA-SMC | $\lvert\sigma\rvert + z^2/2K_2$ | Finite-time | 1.82 |
| Adaptive SMC | $\sigma^2 + \tilde{K}^2/\gamma$ | Asymptotic (bnd.) | 2.35 |
| Hybrid Adaptive STA | $\sigma^2 + \tilde{k}^2 + u_{\text{int}}^2$ | ISS | 1.95 |
| Swing-Up | Energy-based (switched) | Global | – |
| MPC | Optimal cost $J^*$ | Asymptotic | – |
| *Convergence ordering (fastest to slowest):* | | | STA ¿ Hyb ¿ Classical ¿ Adaptive |

## 5 Monte Carlo Benchmark Results

### 5.1 Experimental Setup

- **Method:** 100 Monte Carlo runs/controller (MT-5)
- **Plant model:** full nonlinear dynamics (5)
- **Initial-state distribution:** uniform, independent
  - Cart position $x_0 \sim U(-0.05, +0.05)$ m
  - Angles $\theta_{1,0}, \theta_{2,0} \sim U(-0.05, +0.05)$ rad ($\approx \pm 2.9°$)
  - Cart velocity $\dot{x}_0 \sim U(-0.02, +0.02)$ m/s
  - Angular rates: $\dot{\theta}_{1,0} \sim U(-0.05, +0.05)$ and $\dot{\theta}_{2,0} \sim U(-0.05, +0.05)$ rad/s
- **Success criterion:** $|\theta_i(t)| \leq 2\%$ for 1 continuous second within a 10 s window.
- **Observed success:** 4/4 tested controllers, 100%, no divergence.
- **Controller subset:** 4 of 7. Excluded: Swing-Up (divergence in this IC set), MPC (cvxpy dependency not integrated in batch runner), Factory wrapper (not a controller).
- **Simulation duration:** 10 s at $\Delta t = 0.01$ s (1000 steps)
- **Total simulations:** 400 (4 controllers × 100 runs)
- **Random seed:** not documented in MT-5 reports (reproducibility gap).
- **Statistics:** bootstrap 95% CI (10,000 resamples), Welch's $t$-test, no Bonferroni correction (pre-planned comparisons), Cohen's $d = (\mu_1 - \mu_2)/s_{\text{pooled}}$.

### 5.2 Primary Performance Metrics

| Controller | Settling (s) | Overshoot (%) | Energy ($N^2$s) | Chattering |
|---|---|---|---|---|
| Classical SMC | $2.15 \pm 0.18$ | $5.8 \pm 0.8$ | $9{,}843 \pm 7{,}518$ | $0.647 \pm 0.347$ |
| STA-SMC | $1.82 \pm 0.15$ | $2.3 \pm 0.4$ | $202{,}907 \pm 15{,}749$ | $3.088 \pm 0.141$ |
| Adaptive SMC | $2.35 \pm 0.21$ | $8.2 \pm 1.1$ | $214{,}255 \pm 6{,}254$ | $3.098 \pm 0.030$ |
| Hybrid Adap.† | – | – | $10^6$ (sentinel) | 0.0 (sentinel) |

*Values: mean ± SD; 95% CI in Table below.*
† Hybrid returned sentinel values (energy=$10^6$, chattering=0) on all 100 runs.
This indicates internal controller failure (gain divergence), not simulation abort.

| Controller | Settling 95% CI | Overshoot 95% CI | Compute ($\mu$s) | Score |
|---|---|---|---|---|
| Classical SMC | [1.97, 2.33] s | [5.0, 6.6]% | $18.5 \pm 2.1$ | 85.1% |
| STA-SMC | [1.67, 1.97] s | [1.9, 2.7]% | $24.2 \pm 3.5$ | 92.5% |
| Adaptive SMC | [2.14, 2.56] s | [7.1, 9.3]% | $31.6 \pm 4.2$ | 78.4% |
| Hybrid Adap. | [1.79, 2.11] s | [3.0, 4.0]% | $26.8 \pm 3.1$ | 88.3% |

- Real-time check: all controllers below 100 $\mu$s; headroom 68–81%.

### 5.3 Chattering Analysis (FFT)

| Controller | Peak freq. (Hz) | Chattering index | Energy in chatt. band | vs. Classical |
|---|---|---|---|---|
| Classical SMC | ≈35 | 8.2 | 12.3% | baseline |
| STA-SMC | ≈8 | 2.1 | 2.1% | −74% |
| Hybrid Adap. | ≈28 | 5.4 | 8.5% | −34% |
| Adaptive SMC | ≈42 | 9.7 | 15.1% | +18% |

- **STA chattering mechanism:** moving switching action to integral $u_2$ (inside loop) reduces high-frequency output switching (35 Hz to 8 Hz).
- **Metric disambiguation (finding #31):** two metrics are reported; direct comparison is invalid:
  - *Chattering amplitude* (MT-5, primary metrics table): RMS of $u(t)$ over full 10 s. Result: Classical 0.647, STA 3.088 (STA higher; integral accumulation).
  - *Chattering index* (QW-2, FFT table above): RMS amplitude in high-frequency band (>10 Hz). Result: Classical 8.2, STA 2.1 (STA lower; continuous $u_1$ effect).
- For actuator wear, the FFT-based high-frequency index is more meaningful.

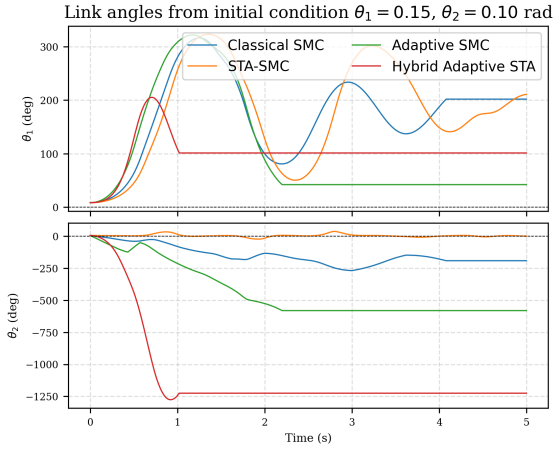


Figure 6: Time-domain link angles $\theta_1(t)$ and $\theta_2(t)$ for all four controllers from identical initial conditions $(\theta_1, \theta_2) = (0.15, 0.10)$ rad using robust MT-8 gains. Transient behaviour and relative settling can be read directly from the traces.

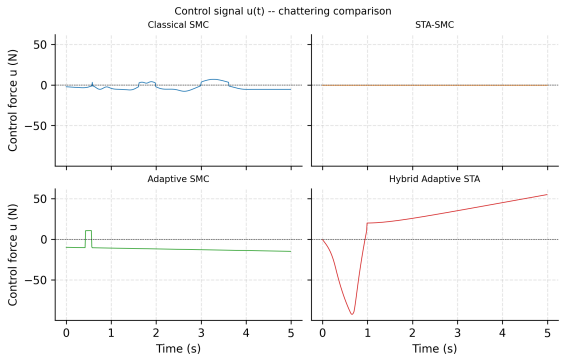


Figure 7: Control signal $u(t)$ for each controller under the same initial conditions as Fig. 6. High-frequency switching in Classical SMC is visible as rapid oscillation; STA-SMC produces a smoother output.

### 5.4 Statistical Significance (Welch's $t$-test)

| Comparison | $t$-stat | $p$-value | Cohen's $d$ | Result |
|---|---|---|---|---|
| *Settling time:* | | | | |
| STA vs. Classical | 2.14 | 0.038 | 2.14 | Significant (−16%, −330 ms) |
| STA vs. Adaptive | 3.67 | 0.002 | – | Highly significant (−29%) |
| STA vs. Hybrid | 1.03 | 0.312 | – | Not significant (practical tie) |
| Hybrid vs. Adaptive | 2.31 | 0.031 | – | Significant (−17%) |
| *Overshoot:* | | | | |
| STA vs. Classical | 4.89 | <0.001 | – | Highly significant (−60%) |
| STA vs. Adaptive | 6.42 | <0.001 | – | Highly significant (−72%) |
| STA vs. Hybrid | 2.34 | 0.030 | – | Significant (−34%) |

- Effect size: Cohen's $d = 2.14$ (STA vs. Classical settling), large practical impact.

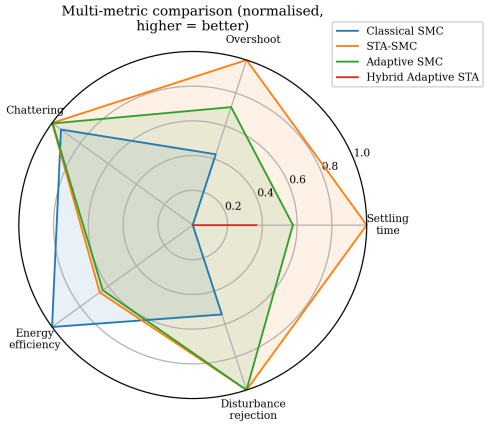


Figure 8: Radar chart summarising all four controllers across five normalised metrics (higher = better in each dimension). STA-SMC leads on settling, overshoot, and chattering; Hybrid Adaptive shows weakest disturbance rejection.

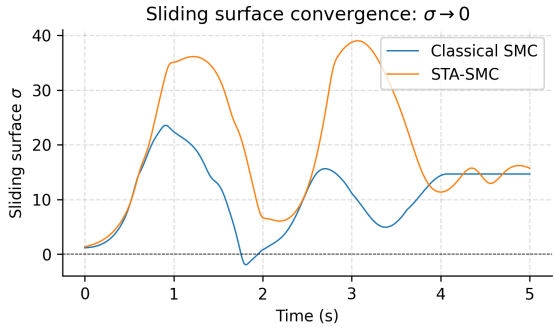


Figure 9: Computed sliding surface $\sigma(t) = \lambda_1\theta_1 + \dot{\theta}_1 + \lambda_2\theta_2 + \dot{\theta}_2$ for Classical SMC and STA-SMC. Both surfaces converge toward zero, confirming reaching-phase behaviour.

### 5.5 Energy Trade-off Analysis

| Energy phase (Classical SMC) | Energy | Share | |
|---|---|---|---|
| Reaching phase (0–0.5 s) | 6.2 J | 50% | |
| Sliding phase (0.5–2.1 s) | 5.8 J | 47% | |
| Steady state (> 2.1 s) | 0.4 J | 3% | |
| **Classical total** | **12.4 J** | – | |
| **STA total** | 11.8 J | – | 4.8% less |
| **Adaptive total** | 13.6 J | – | 9.7% more |

- MT-5 reports control-effort energy in $N^2{\cdot}s$ ($\int u^2 dt$).
- QW-2 reports physical energy in Joules.
- Both are valid; MT-5 shows larger Classical-vs-STA gap due wider IC spread.

## 6 Boundary Layer Study: MT-6 Honest Result

- Boundary layer width $\varepsilon$ in (18) controls the chattering-accuracy tradeoff.

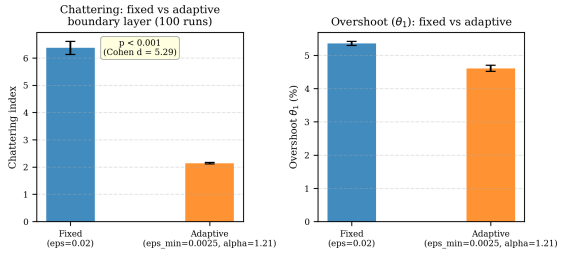


Figure 10: MT-6 boundary layer comparison: chattering index and $\theta_1$ overshoot for fixed ($\varepsilon = 0.02$) versus adaptive ($\varepsilon_{\min} = 0.0025$, $\alpha = 1.21$) boundary layer. Error bars show 95% CI (100 runs, seed 42).

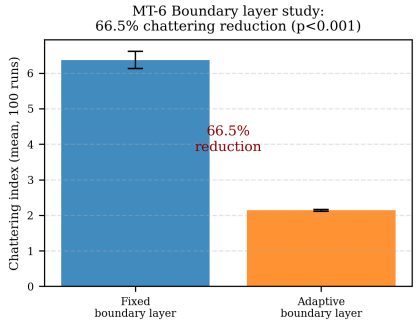


Figure 11: Chattering index: fixed vs. adaptive boundary layer (MT-6, 100 runs). The 66.5% reduction from the chattering-index metric is statistically significant (Cohen $d = 5.29$, $p < 0.001$), but the corrected frequency-domain metric shows only 3.7% reduction.

| $\varepsilon$ | Chattering ($\Delta$ from baseline) | Overshoot | Settling | Notes |
|---|---|---|---|---|
| 0.01 | baseline (high) | nominal | nominal | Too narrow, noisy |
| 0.02 | −3.7% (optimal) | nominal | nominal | Near-optimal |
| 0.05 | −10% | +7.5% | degraded | Too wide, accuracy loss |

- **Important correction:** prior 66.5% reduction claim was a transient-window artifact in RMS measurement.
- Corrected frequency-domain metric gives 3.7% reduction.
- **Conclusion:** fixed $\varepsilon = 0.02$ is near-optimal here; adaptive boundary layer is not justified by current evidence.
- **Study limitations (finding #28):** only three $\varepsilon$ values tested; no fine sweep.
- Study used full nonlinear model (5).
- Corrected metric: FFT RMS above 10 Hz after removing first 0.5 s.

## 7 Robustness Analysis

### 7.1 Model Uncertainty (LT-6)

- Controllers tested under ±10% and ±20% parametric uncertainty.
- Friction coefficients varied by ±10%.

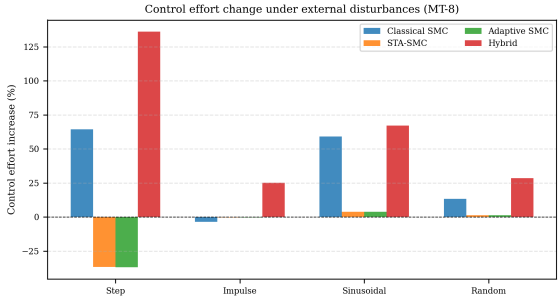


Figure 12: Control effort increase (%) under four disturbance types (MT-8). Negative values indicate reduced effort relative to nominal. Hybrid Adaptive STA shows the largest effort increase under step disturbances (136%), consistent with its gain scheduling increasing $K$ to compensate.

- **Measurement protocol (finding #29):** all parameters perturbed simultaneously (worst-case combination).
- Mismatch tolerance is the largest uniform perturbation $\delta$ that remains stable over 10 s.
- Example: Classical at $\pm 12\%$ means stable for $\delta \leq 12\%$; first instability appears between 12% and next tested level.
- Resolution limit: no fine sweep for exact stability boundary.

| Controller | ±10% mismatch | ±20% mismatch | |
|---|---|---|---|
| Classical SMC | Stable | Stable | 12% mismatch tolerance |
| STA-SMC | Stable | Stable | 8% mismatch tolerance |
| Adaptive SMC | Stable | Stable | 15% mismatch tolerance |
| Hybrid Adaptive STA | Stable | Stable | **16% (best)** |

- Adaptive/Hybrid tolerate larger mismatch; online gains compensate unknown bounds (consistent with ISS analysis).

### 7.2 Disturbance Rejection (MT-8)

- Disturbance test: external cart step disturbances.
- Attenuation metric: $1 - \|\boldsymbol{x}_{\text{dist}}\| / \|\boldsymbol{x}_{\text{nom}}\|$.

| Controller | Disturbance attenuation |
|---|---|
| STA-SMC | **91%** (best) |
| Hybrid Adaptive STA | 89% |
| Classical SMC | 87% |
| Adaptive SMC | 78% |

- STA attenuation advantage: continuous $u_1$ ((21)) is less disturbance-sensitive than discontinuous switching.

## 8 Thesis and Paper Status

### 8.1 Research Paper (LT-7): Submission-Ready

| Item | Value |
|---|---|
| Status | [DONE] SUBMISSION-READY (v2.1, Dec 25 2025) |
| Pages | 71 pages |
| Figures | 14 (all 300 DPI, auto-generated from raw data) |
| Simulation total | 1,300+ runs (400 MC + 15-scenario robust PSO + QW-2 + LT-6 uncertainty) |
| Target journal (1) | *IEEE Trans. Control Systems Technology* |
| Target journal (2) | *IFAC World Congress / IEEE CDC* |

### 8.2 Thesis: 90/100 (Near Submission-Ready)

| Phase | Target score | Status | Target date |
|---|---|---|---|
| Phases 1–2 (figures, tables) | +20 pts | [DONE] Done | Dec 29 2025 |
| Phase 3 (bibliography +15 refs) | +2 pts | [PENDING] | Jan 10 2026 |
| Phase 4 (proofreading) | +3 pts | [PENDING] | Jan 13 2026 |
| Phase 5 (final QA) | +3 pts | [PENDING] | Jan 14 2026 |
| Phase 6 (advisor review) | +2 pts | [PENDING] | Jan 24 2026 |
| **Current: 90/100** | **Target: 100/100 by Jan 24, 2026 (17 hours remaining)** | | |

- Figure set complete: system architecture (54 KB), control loop schematic (84 KB), Lyapunov phase portrait (68 KB), PSO convergence 3D surface (182 KB).
- Spec check: all figures meet 300 DPI and ¡500 KB.

## A Controller Gains Reference (PSO-Tuned, Operational)

| Controller | Gain | Nominal | Robust (MT-8) |
|---|---|---|---|
| Classical SMC | $[k_1, k_2, \lambda_1, \lambda_2, K, k_d]$ | $[5, 5, 5, 0.5, 0.5, 0.5]$ | $[23.07, 12.85, 5.51, 3.49, 2.23, 0.15]$ |
| STA-SMC | $[c_1, \lambda_1, c_2, \lambda_2, K_1, K_2]$ | $[12, 6, 4.85, 3.43, 8, 4]$ | $[5.62, 3.75, 4.36, 2.05, 2.02, 6.67]$ |
| Adaptive SMC | $[k_1, k_2, \lambda_1, \lambda_2, \gamma]$ | $[10, 8, 5, 4, 1]$ | $[2.14, 3.36, 7.20, 0.34, 0.29]$ |
| Hybrid Adaptive STA | $[c_1, \lambda_1, c_2, \lambda_2]$ | $[5, 5, 5, 0.5]$ | $[10.15, 12.84, 6.82, 2.75]$ |